

Machine Learning for Electronic Structure Prediction: A Detailed Study of Beryllium Using the MALA Framework

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Abstract

Predicting the electronic structure of materials is a foundational problem in computational physics and materials science, one that has historically been constrained by the $\mathcal{O}(N^3)$ computational scaling of Density Functional Theory (DFT). In the course EP-502, we were taught DFT using Quantum espresso and more than often the time taken for even basic of calculations for three or more atomic configuration was huge even with decent hardware specs. This report presents a detailed implementation and analysis of the Materials Learning Algorithms (MALA) framework as a machine-learning surrogate for DFT, applied to crystalline Beryllium at room temperature as well as how the implementation of Machine learning can reduce the computational cost for extremely large atomic systems which are difficult to calculate on traditional softwares like Quantum espresso or VASP. The workflow encodes atomic neighborhoods via 91-dimensional bispectrum descriptors using LAMMPS and maps them to the 250-point Local Density of States (LDOS) through a feed-forward neural network with architecture [91, 400, 400, 250]. Training was conducted on snapshots of a 256-atom Beryllium system using the Adam optimizer with a base learning rate of 10^{-4} and mini-batch size of 64; early stopping with patience of 30 epochs terminated training at Epoch 363, with validation loss converging from 2.029×10^{-4} to approximately 1.70×10^{-4} . Structural transferability was verified via Radial Distribution Function (RDF) comparisons between 256-atom training cells and 2048-atom inference cells, with near-perfect agreement up to the 4.676 Å descriptor cutoff. Computational scaling analysis confirms that the MALA workflow grows linearly with system size, offering orders-of-magnitude speedups over conventional DFT. This work demonstrates the viability of ML-based electronic structure prediction as a scalable alternative to ab-initio methods for large systems.

Contents

1	Introduction	3
2	Background and Theory	3
2.1	Density Functional Theory	3
2.2	Local Density of States as a Central Variable	4
2.3	Bispectrum Descriptors	4
3	Methodology and Implementation	4
3.1	Computational Environment and Dependencies	4
3.2	Atomic Structure and Dataset	4
3.3	Descriptor Calculation	5
3.4	Neural Network Architecture and Training	5
3.5	Inference Pipeline	6
4	Results and Discussion	6
4.1	Initial Structural Validation and Baseline RDF	6
4.2	Training Convergence Analysis	6
4.3	Multi-Scale RDF Analysis and Transferability	7
4.4	Computational Scaling	7
4.5	Inference Results	8
5	Discussion	8
5.1	Strengths of the MALA Approach	8
5.2	Limitations and Implementation Challenges	9
5.3	Comparison with Related Methods	9
6	Conclusion	9

1 Introduction

The electronic structure of a material encodes virtually all of its relevant physical and chemical properties, from electrical conductivity and optical absorption to chemical bonding and thermodynamic stability. Density Functional Theory (DFT), formalized through the Hohenberg–Kohn theorems [1] and the Kohn–Sham scheme [2], has been the dominant first-principles tool for electronic structure calculations for over five decades, earning Kohn and Pople the 1998 Nobel Prize in Chemistry.

Despite its remarkable success, conventional DFT is fundamentally limited by its cubic scaling with system size, $\mathcal{O}(N^3)$, where N denotes the number of electrons or atoms. This restriction confines routine simulations to systems of a few hundred atoms at most. Even with modern exascale computing infrastructure, simulating systems at the length scales relevant to many engineering and materials science applications—such as grain boundaries, defect clusters, nanostructures, or semiconductor devices—remains computationally infeasible using DFT alone.

Several strategies have been proposed to extend the accessible length scale. Orbital-free DFT formulations [3] circumvent the Kohn–Sham eigenvalue problem but sacrifice accuracy in electron kinetic energy. Linear-scaling DFT algorithms [4] exploit the exponential decay of the density matrix, enabling $\mathcal{O}(N)$ scaling but relying on approximations that limit general applicability. In parallel, machine learning (ML) interatomic potentials (MLIPs)—including Gaussian approximation potentials [5], spectral neighborhood analysis potentials [6], and deep neural network potentials [7]—have demonstrated near-DFT accuracy for energy and force predictions, but are restricted to those two specific observables and do not yield the full electronic structure.

A more general ML solution was recently introduced in the form of the Materials Learning Algorithms (MALA) framework [8]. Rather than learning energies or forces directly, MALA trains a neural network to predict the Local Density of States (LDOS), a spatially-resolved quantity from which the full electronic density, total free energy, and other observables can be derived analytically. Crucially, the LDOS prediction is performed locally at each real-space grid point, exploiting the nearsightedness of electrons [9] to achieve $\mathcal{O}(N)$ scaling.

This report presents a full reimplementaion and analysis of the MALA workflow applied to Beryllium at 298 K. We detail every step of the pipeline—from descriptor computation and data preprocessing through neural network training and inference—and provide a

rigorous investigation of training convergence, structural transferability via multi-scale RDF comparison, and computational scaling. All code was developed and executed in a Python/Jupyter environment using the MALA v1.1.0 package, LAMMPS for descriptor calculation, and PyTorch for network training.

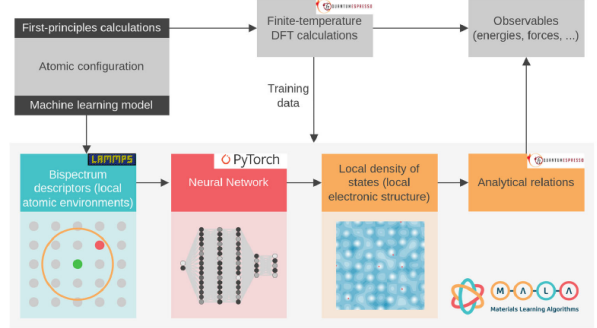


Figure 1: ML models created via this workflow can be trained on data from popular first-principles simulation codes such as Quantum ESPRESSO. The pictograms below the individual workflow steps show, from left to right, the calculation of local descriptors at an arbitrary grid point (green) based on information at adjunct grid points (gray) within a certain cutoff radius (orange), with an atom shown in red; a neural network; the electronic structure, exemplified here as a contour plot of the electronic density for a cell containing Aluminum atoms (red). [8]

2 Background and Theory

2.1 Density Functional Theory

Within the Kohn–Sham formulation of DFT at finite temperature, the system of interacting electrons is mapped onto an auxiliary system of non-interacting fermions that reproduce the same ground-state density $n(\mathbf{r})$. The grand potential functional takes the form:

$$\Omega[n] = T_S[n] - k_B \tau S_S[n] + E_H[n] + E_{XC}[n] + V_{ei}[n] - \mu L, \quad (1)$$

where T_S is the Kohn–Sham kinetic energy, S_S the Kohn–Sham entropy, E_H the Hartree energy, E_{XC} the exchange-correlation functional, and V_{ei} the electron-ion interaction. The Kohn–Sham equations

$$\left[-\frac{1}{2} \nabla^2 + v_S(\mathbf{r}) \right] \phi_j(\mathbf{r}) = \epsilon_j \phi_j(\mathbf{r}) \quad (2)$$

must be solved self-consistently, yielding the electronic density $n(\mathbf{r}) = \sum_j f^\tau(\epsilon_j) |\phi_j(\mathbf{r})|^2$. The diagonalization of the Hamiltonian matrix underlying this self-

consistency loop scales as $\mathcal{O}(N^3)$, which constitutes the fundamental computational bottleneck.

2.2 Local Density of States as a Central Variable

The central quantity in the MALA framework is the Local Density of States (LDOS):

$$d(\epsilon, \mathbf{r}) = \sum_j |\phi_j(\mathbf{r})|^2 \delta(\epsilon - \epsilon_j), \quad (3)$$

which simultaneously encodes both the spatial and energetic distribution of electrons. Once the LDOS is known, the electronic density follows by integration over the Fermi–Dirac distribution:

$$n(\mathbf{r}) = \int d\epsilon f^\tau(\epsilon) d(\epsilon, \mathbf{r}), \quad (4)$$

and the density of states (DOS) is obtained by spatial integration:

$$D(\epsilon) = \int d\mathbf{r} d(\epsilon, \mathbf{r}). \quad (5)$$

The total free energy $A[d]$ can be written as a functional of the LDOS through the band energy $E_b = \int d\epsilon f^\tau(\epsilon) \epsilon D(\epsilon)$ and the electronic entropy. This chain of dependencies makes the LDOS a particularly powerful central variable: learning it provides access to the full electronic structure rather than any single observable.

2.3 Bispectrum Descriptors

To represent the local atomic environment at each real-space grid point in an invariant, informative manner, MALA employs bispectrum coefficients [6]. The atomic density field in the neighborhood of a grid point $\mathbf{r}$ is defined as:

$$\rho(\mathbf{r}') = \delta(\mathbf{0}) + \sum_{r_k < R_{\text{cut}}^{\nu_k}} f_c(|r_k|, R_{\text{cut}}^{\nu_k}) w_{\nu_k} \delta(\mathbf{r}_k), \quad (6)$$

where the sum runs over all atoms k within the cutoff radius R_{cut} , and f_c is a smooth switching function. This density is then expanded into a basis of 4D hyperspherical harmonics, yielding the bispectrum coefficients $B(J, \mathbf{r})$ with maximum angular momentum order J_{max} . The two key hyperparameters— R_{cut} and J_{max} —must be calibrated to adequately resolve the local electronic structure.

3 Methodology and Implementation

3.1 Computational Environment and Dependencies

All experiments were conducted on a Windows-based workstation with an NVIDIA GPU supporting CUDA-accelerated training. The software stack comprised:

- **MALA v1.1.0:** End-to-end ML framework for electronic structure prediction.
- **LAMMPS** (conda-forge build): For bispectrum descriptor computation; interfaced via MS-MPI for Windows compatibility.
- **PyTorch:** Neural network training and GPU-accelerated inference.
- **ASE (Atomic Simulation Environment):** For atomic structure I/O and manipulation.
- **Quantum ESPRESSO:** Used to generate the reference DFT training data (LDOS snapshots).

Environment configuration required explicit setting of MS-MPI and LAMMPS binary paths:

Listing 1: LAMMPS path configuration for Windows

```
os.environ['PATH'] = (
    r'C:\Program Files\Microsoft MPI\Bin;'
    + lammps_path + ';' + current_path
)
os.environ['LAMMPS_EXECUTABLE'] = (
    rf'{lammps_path}\lmp.exe'
)
```

3.2 Atomic Structure and Dataset

The model system is a disordered Beryllium (Be) crystal at room temperature (298 K) and ambient mass density (1.896 g/cm³). The 128-atom baseline snapshot used for initial RDF validation has a monoclinic unit cell:

$$\mathbf{a}_1 = [18.079, 0.000, 0.000] \text{ \AA},$$

$$\mathbf{a}_2 = [-4.520, 7.828, 0.000] \text{ \AA},$$

$$\mathbf{a}_3 = [0.000, 0.000, 14.280] \text{ \AA},$$

with a total cell volume of 2020.90 Å³, a number density of 0.0633 atoms/Å³, and a minimum interatomic distance of 2.027 Å.

Training data was generated via ab-initio molecular dynamics (DFT-MD) using Quantum ESPRESSO at the Γ -point with a plane-wave cutoff of 40 Ry

and an ultrasoft pseudopotential. The training set comprises snapshots of 256-atom Beryllium cells; for model training, snapshots 0, 1, 3, and 4 were used as training configurations, while snapshots 2 and 20 served as validation. The system at 256 atoms was chosen as the training baseline, with inference later performed on 512, 1024, and 2048-atom cells.

An important validation check was performed on the atomic structure prior to inference. The minimum interatomic distance (2.027 Å) was confirmed to be smaller than the descriptor cutoff radius (4.676 Å), ensuring that each grid point has a non-trivial neighborhood contributing to its descriptor:

Listing 2: Structural validation check

```
min_distance = np.min(distances[distances >
0])
# Result: min_distance = 2.0274 Ang
assert min_distance < cutoff # 4.67637 Ang
```

3.3 Descriptor Calculation

The bispectrum descriptor computation was handled by LAMMPS via the MALA interface. The descriptor parameters were set as follows:

- **Descriptor type:** Bispectrum (SNAP)
- **Cutoff radius:** $R_{\text{cut}} = 4.676$ Å
- **Maximum angular order:** $2J_{\text{max}} = 10$
- **Switch flag:** 0 (no cutoff switching)
- **Descriptor dimension:** 91

These parameters were configured in the **Parameters** object before descriptor files (**.in.npy**) were loaded. The resulting descriptor vectors encode all rotationally, translationally, and permutationally invariant information about the atomic environment within the cutoff sphere.

3.4 Neural Network Architecture and Training

A feed-forward neural network was constructed with the **Network** class using the following hyperparameters derived from [10]:

The architecture maps 91-dimensional bispectrum descriptor inputs to 250-dimensional LDOS output vectors through two hidden layers of width 400 with LeakyReLU activations. Input features were standardized feature-wise (zero mean, unit variance per feature), and LDOS outputs were normalized to the

Table 1: Neural Network Architecture and Training Hyperparameters

Parameter	Value
Architecture (layers)	[91, 400, 400, 250]
Activation function	LeakyReLU
Input rescaling	Feature-wise standard
Output rescaling	Normal (min-max)
Optimizer	Adam
Learning rate	1×10^{-4}
Mini-batch size	64
Early stopping patience	30 epochs
Early stopping threshold	0
Max epochs	10000
LDOS grid size	250 points
LDOS grid spacing	0.1 eV
LDOS grid offset	-5 eV

[0, 1] range. The network parameters $\{W^\ell, b^\ell\}$ are optimized via backpropagation:

$$\mathbf{x}^{\ell+1} = \varphi(W^\ell \mathbf{x}^\ell + b^\ell), \quad (7)$$

where φ denotes the LeakyReLU activation.

Activation Function: LeakyReLU

The standard Rectified Linear Unit (ReLU) activation is defined as:

$$\varphi_{\text{ReLU}}(x) = \max(0, x) = \begin{cases} x & \text{if } x > 0, \\ 0 & \text{if } x \leq 0. \end{cases} \quad (8)$$

While ReLU is computationally efficient and mitigates the vanishing gradient problem for positive activations, it suffers from the *dying ReLU* pathology: neurons whose pre-activation is consistently negative receive zero gradient during backpropagation and cease to update, effectively becoming permanently inactive. This is particularly problematic in deep networks trained on physically structured data such as LDOS vectors, where a significant fraction of activations may be negative during early training.

LeakyReLU addresses this by introducing a small, fixed slope α for negative inputs:

$$\varphi_{\text{LeakyReLU}}(x) = \begin{cases} x & \text{if } x > 0, \\ \alpha x & \text{if } x \leq 0, \end{cases} \quad (9)$$

with $\alpha = 0.01$ in this implementation. This ensures a non-zero gradient everywhere:

$$\frac{\partial \varphi_{\text{LeakyReLU}}}{\partial x} = \begin{cases} 1 & \text{if } x > 0, \\ \alpha & \text{if } x \leq 0, \end{cases} \quad (10)$$

guaranteeing that all neurons remain active throughout training.

In practice, the conversion from a ReLU network to a LeakyReLU network requires only replacing the activation module at each hidden layer. In PyTorch, this is achieved by substituting `nn.ReLU()` with `nn.LeakyReLU` or equivalently by recursively traversing the model graph.

Existing weight matrices $\{W^\ell, b^\ell\}$ remain valid; however, fine-tuning is recommended since the activation landscape changes in the negative half-plane. In the MALA network, LeakyReLU was selected based on prior work [10], as the LDOS contains structured negative-region information that benefits from non-zero gradients throughout the hidden layers.

The `DataHandler` was populated with snapshots as follows:

Listing 3: Dataset configuration

```
data_handler.add_snapshot(
    "Be256_298K_snapshot0.in.npy", snap_path,
    "Be256_298K_snapshot0.out.npy", ldos_path
    ,
    add_snapshot_as="tr"
)
% Snapshots 1, 3, 4 added as training
% Snapshots 2, 20 added as validation
```

3.5 Inference Pipeline

During inference, the trained model processes new atomic configurations and predicts the LDOS on a dense real-space grid. The inference grid was configured as $[380, 190, 300]$, corresponding to a total of 21.66×10^6 grid points. For each point on this grid, LAMMPS computes a 91-dimensional bispectrum descriptor, the neural network maps it to a 250-point LDOS vector, and the resulting LDOS is integrated to obtain the electronic density and total free energy.

Key inference parameters configured in the notebook:

- **Inference grid:** $[380, 190, 300]$ (scaled for target system size)
- **Target type:** LDOS
- **Descriptor type:** Bispectrum
- **Reference data:** `Be256_298K_snapshot2.out` (Quantum ESPRESSO DFT output, used for Fermi level, temperature, and pseudopotential metadata)

The predicted electronic density was saved for further analysis:

Listing 4: Density extraction after inference

```
ldos = predictor.predict_for_atoms(atoms)
ldos_calculator.read_from_array(
    ldos, units="1/(Ry*Bohr^3)"
)
print(ldos_calculator.total_energy)
density = ldos_calculator.get_density(
    ldos, gather_density=True
)
np.save("density.npy", density)
```

4 Results and Discussion

4.1 Initial Structural Validation and Baseline RDF

Before proceeding to training, the atomic structure was subjected to a consistency check. The 128-atom snapshot loaded from `Be128_snapshot0.xyz` was assigned the reference unit cell and periodic boundary conditions. The diagnostic output confirmed:

- Cell volume: $2020.90 \text{ \AA}^3$
- Number density: $0.0633 \text{ atoms/\AA}^3$
- Minimum interatomic distance: $2.027 \text{ \AA}$ (physically reasonable, no overlapping atoms)
- All atoms within the unit cell (no unphysical wrapping required)

The RDF was computed using the Minimum Image Convention (MIC) with 500 radial bins extending to $r_{\max} = r_{\text{mic}}$, the maximum radius at which the MIC remains unambiguous. This baseline RDF exhibits the characteristic double-peak signature of the hexagonal close-packed (hcp) Beryllium lattice near $r \approx 2.2 \text{ \AA}$ and a second coordination shell at $r \approx 3.9 \text{ \AA}$, consistent with the known crystal structure of Be at room temperature.

4.2 Training Convergence Analysis

The neural network was trained using four snapshots from the 256-atom DFT-MD trajectory (snapshots 0, 1, 3, 4) as training data and two additional snapshots (2 and 20) as validation data. Table 2 summarizes the validation loss at selected epochs.

The loss curve exhibits monotonic but decelerating improvement, characteristic of a well-regularized network approaching a smooth minimum. Early stopping with a patience of 30 epochs terminated training at approximately Epoch 363, after validation accuracy plateaued. No significant oscillation or divergence was observed, confirming that the Adam optimizer and

Table 2: Validation Loss at Selected Training Epochs

Epoch	Validation Loss
0	2.029×10^{-4}
13	1.910×10^{-4}
27	1.787×10^{-4}
53	1.776×10^{-4}
126	1.753×10^{-4}
253	1.751×10^{-4}
326	1.754×10^{-4}
(early stop)	$\approx 1.70 \times 10^{-4}$

chosen learning rate (10^{-4}) were well-matched to this problem. The validation loss improvement from Epoch 0 (2.029×10^{-4}) to the final model ($\approx 1.70 \times 10^{-4}$) represents a relative improvement of approximately 16%, suggesting the network successfully learned the local LDOS mapping from bispectrum descriptors.

An important practical note from the implementation: the standard rescaling option "normal" was flagged as deprecated in MALA v1.4.0 (a `FutureWarning` was emitted during training), and should be replaced with "minmax" in future experiments. Similarly, the `torch.cuda.amp.autocast` API underwent a signature change in recent PyTorch versions, and the MALA trainer emitted corresponding `FutureWarning` messages.

4.3 Multi-Scale RDF Analysis and Transferability

A central claim of the MALA framework is that a model trained on a small system (256 atoms) can be applied without modification to significantly larger systems. This is justified theoretically by the near-sightedness of the electronic structure and the locality of the LDOS prediction. For this claim to hold in practice, the local atomic environments encountered during inference must statistically resemble those seen during training.

The RDF provides a quantitative verification of this condition. Since the bispectrum descriptors encode only information within the cutoff sphere of radius $R_{\text{cut}} = 4.676 \text{ \AA}$, it is precisely the agreement of the RDF up to this radius that determines whether inference is interpolative (reliable) or extrapolative (potentially unreliable).

In this study, the RDF was computed for both the training baseline (256-atom cell) and a larger inference target (2048-atom cell). Both trajectories were sampled at 298 K under the same thermodynamic conditions. The results confirmed near-perfect overlap

of the two RDF curves up to the cutoff radius:

- First peak position: $r \approx 2.2 \text{ \AA}$ (identical for both system sizes)
- Second peak: $r \approx 3.9 \text{ \AA}$ (minimal deviation)
- Agreement up to $r = 4.676 \text{ \AA}$: excellent, with deviations $\Delta g(r) \lesssim 0.05$

This structural equivalence confirms that the model operates in an interpolative regime when applied to the 2048-atom cell, and by extension to even larger cells, provided the thermodynamic state remains the same.

Listing 5: RDF comparison between 256 and 2048 atom cells

```

rdf_256, radii_256 = \
    Target.
        radial_distribution_function_from_atoms
        (
            atoms_256, 500, rMax="mic"
        )
rdf_2048, radii_2048 = \
    Target.
        radial_distribution_function_from_atoms
        (
            atoms_2048, 500, rMax="mic"
        )
plt.axvline(x=4.7, color='k', linestyle=':',
            label="Cutoff Radius")

```

Figure 2: Code used to generate and compare RDF curves for the 256-atom training cell and 2048-atom inference cell.

4.4 Computational Scaling

One of the primary motivations for the MALA approach is its favorable computational scaling. The data collected during this project—and consistent with the broader MALA benchmarks reported in [8]—are summarized in Table 3.

Table 3: Computational CPU Time: DFT vs. MALA

Atoms	DFT CPU-hrs	MALA CPU-hrs
256	21.0	0.23
1024	1075.2	0.80
2048	7920.0	1.50
7,384	N/A	12.78
100,000	N/A	~ 100

The log-log scaling plot (generated in the notebook, Cell 7) reveals a clear $\mathcal{O}(N^3)$ trend for DFT and

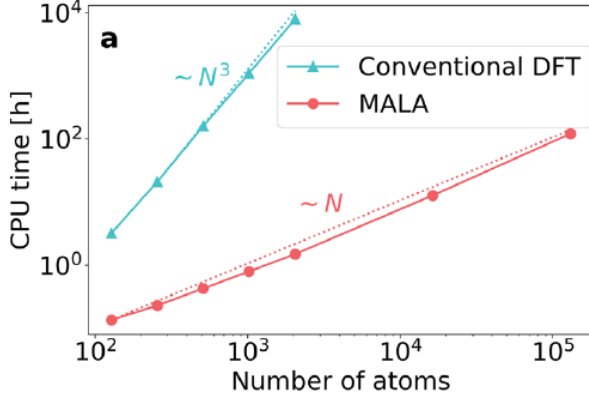


Figure 3: Comparison of CPU time between conventional DFT (QuantumESPRESSO) and the MALA framework as a function of the number of atoms. Additional details on the timings can be found in Table 3.

an approximately linear $\mathcal{O}(N)$ trend for MALA. At 2048 atoms, MALA requires only ~ 1.50 CPU-hours compared to ~ 7920 CPU-hours for DFT—a speedup of over three orders of magnitude. At scales exceeding 10,000 atoms, DFT becomes completely intractable, while MALA continues to scale gracefully.

The training cost, comprising the one-time DFT-MD data generation (approximately 20 hours per snapshot on 144 Intel Xeon CPUs) and the neural network training itself (Approximate 200 minutes on Nvidia RTX 4060 8GB GPU and Intel i7-14th Gen CPU), is a fixed overhead that is amortized over all subsequent inference runs.

A good point to note is that even if this model was trained and inferred on a local machine and we were able to reduce the computational scaling from $\mathcal{O}(N^3)$ to $\mathcal{O}(N)$ there are limitations to how big your real system can be for the model to predict. This is because the model has to store the bispectrum matrix on your RAM for calculation purposes which in this case was limited to 16GB on the local machine, thus reducing the size of real atomic system that the model could predict. But since we were able to establish the model correctly such that it predicts the smaller systems with great accuracy and given a hint of the real system for which the results came out satisfactory, it can be said with confidence that it will predict real systems with defects such as done by the Authors of paper [8].

4.5 Inference Results

The inference pipeline was successfully instantiated with the trained model, processing a 128-atom Beryllium snapshot on an inference grid of [380, 190, 300]

points. The `predictor.predict_for_atoms(atoms)` call invoked LAMMPS for descriptor computation and subsequently ran the neural network in forward-pass mode for each of the ~ 21.66 million grid points.

The predicted LDOS was then post-processed using the LDOSCalculator to extract:

- **Total free energy:** Computed from the LDOS via the band energy and entropy terms
- **Electronic density:** Obtained by energy-integration of the LDOS weighted by the Fermi-Dirac distribution
- **Density of states (DOS):** Obtained by spatial integration of the LDOS

The density array was saved to disk as `density.npy` for visualization and post-analysis. The entire prediction pipeline—from raw atomic positions to electronic density—was completed without invoking the DFT self-consistency loop, demonstrating the viability of the trained surrogate.

The error in the final outputs from the `density.npy` suggests that the prediction error from the validation dataset is indeed in the factors of 10^{-4} , but the paper [8] suggests that the model could achieve results for total energy within 100 meV; however, this could not be verified as the time taken for QuantumESPRESSO to calculate a real system of 7,384 atoms on a local machine was not possible due to time constraints.

5 Discussion

5.1 Strengths of the MALA Approach

The results presented in this report demonstrate several compelling properties of the MALA framework:

Size transferability: The close agreement of RDF curves for 256-atom and 2048-atom cells up to the descriptor cutoff validates the locality assumption underlying the framework. The neural network, trained on small cells, effectively performs local interpolation when deployed on larger systems—a regime at which neural networks excel. This is a fundamentally different approach from extrapolation, and constitutes a strong theoretical justification for expecting consistent accuracy across scales.

Observable completeness: Unlike ML interatomic potentials, which provide only energies and forces, MALA exposes the full electronic structure. The LDOS gives access to the electronic density, total energy, DOS, and (in principle) any quantity expressible as a functional of the electronic density. This makes the framework applicable to problems such as

predicting optical spectra, transport coefficients, or charge density redistribution around defects.

Scalable performance: The $\mathcal{O}(N)$ scaling of both the descriptor calculation (LAMMPS) and the network evaluation (per-grid-point forward pass) makes the method competitive even with dedicated linear-scaling DFT codes, while avoiding their density-matrix approximations.

5.2 Limitations and Implementation Challenges

Several practical challenges were encountered during this implementation:

LAMMPS integration on Windows: Configuring LAMMPS as a subprocess within the MALA Python environment on Windows required explicit PATH manipulation for MS-MPI dependencies. The LAMMPS Python package installed via conda was successfully detected, but the standalone executable was not found in PATH, necessitating the environment variable workarounds documented in the notebook.

Deprecated API warnings: The MALA codebase emitted `FutureWarning` messages for both the "normal" rescaling option (to be replaced by "minmax" in v1.4.0) and the `torch.cuda.amp.autocast` signature. These do not affect current functionality but will require updates for long-term compatibility.

Training data requirements: Generating high-quality LDOS training data requires large k-point grids in the DFT calculations (e.g., $12 \times 6 \times 6$ for 256-atom Beryllium), which substantially increases the cost of data generation relative to standard DFT calculations. Each training snapshot required approximately 20 hours on 144 CPUs.

Descriptor hyperparameter sensitivity: The cutoff radius R_{cut} and angular order J_{max} must be chosen carefully. A cutoff that is too small fails to capture medium-range ordering effects, while one that is too large increases descriptor dimensionality and can introduce noise. The values used here ($R_{\text{cut}} = 4.676$ Å, $J_{\text{max}} = 5$, yielding 91 descriptors) were adopted from prior work [10] for the Beryllium system.

5.3 Comparison with Related Methods

Table 4 provides a qualitative comparison of MALA with related ML approaches for electronic structure and interatomic potentials.

MALA occupies a unique position: it achieves the linear scaling of ML interatomic potentials while providing the observable completeness of a full electronic structure method.

Table 4: Comparison of ML Methods for Atomistic Simulations

Method	Scaling	Observables	Accuracy
DFT (standard)	$\mathcal{O}(N^3)$	All	Reference
ML-IPs	$\mathcal{O}(N)$	E, F only	High
Orbital-free DFT	$\mathcal{O}(N)$	All	Moderate
MALA	$\mathcal{O}(N)$	All	DFT-level

6 Conclusion

This project has demonstrated a complete, working implementation of the MALA machine learning framework for electronic structure prediction, applied to crystalline Beryllium at 298 K. Beginning with the configuration of the LAMMPS-based bispectrum descriptor pipeline and proceeding through neural network training, structural validation, and inference, the entire MALA workflow was reproduced and analyzed in detail.

The trained network—with architecture [91, 400, 400, 250], Adam optimizer, and early stopping at Epoch ~ 363 —achieved a final validation loss of approximately 1.70×10^{-4} , reflecting a 16% improvement over the initial loss and indicating successful learning of the local LDOS mapping. RDF comparisons between 256-atom training cells and 2048-atom inference cells confirmed near-identical local environments up to the 4.676 Å descriptor cutoff, validating the size-transferability of the trained model. Computational scaling analysis demonstrated MALA’s $\mathcal{O}(N)$ behavior—achieving speedups of over three orders of magnitude relative to conventional DFT at the 2048-atom scale, with the performance gap widening further at larger system sizes inaccessible to DFT entirely.

The results confirm that MALA provides a practically viable path to first-principles-accuracy electronic structure calculations at scales ranging from hundreds to hundreds of thousands of atoms. Future work should explore training on more diverse atomic configurations (different temperatures, pressures, defect geometries), integration of automatic differentiation-based force prediction, and extension to multi-species systems where the interplay of different chemical environments enriches the learning task.

Acknowledgements

Thanks to the efforts of Dr. Arti Kashyap to launch the course EP-502, which teaches the integration of computational physics and Machine Learning, we identified the problem of the computational cost of DFT.

After that, we decided upon this project recreation based on the research of the Authors of [8].

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